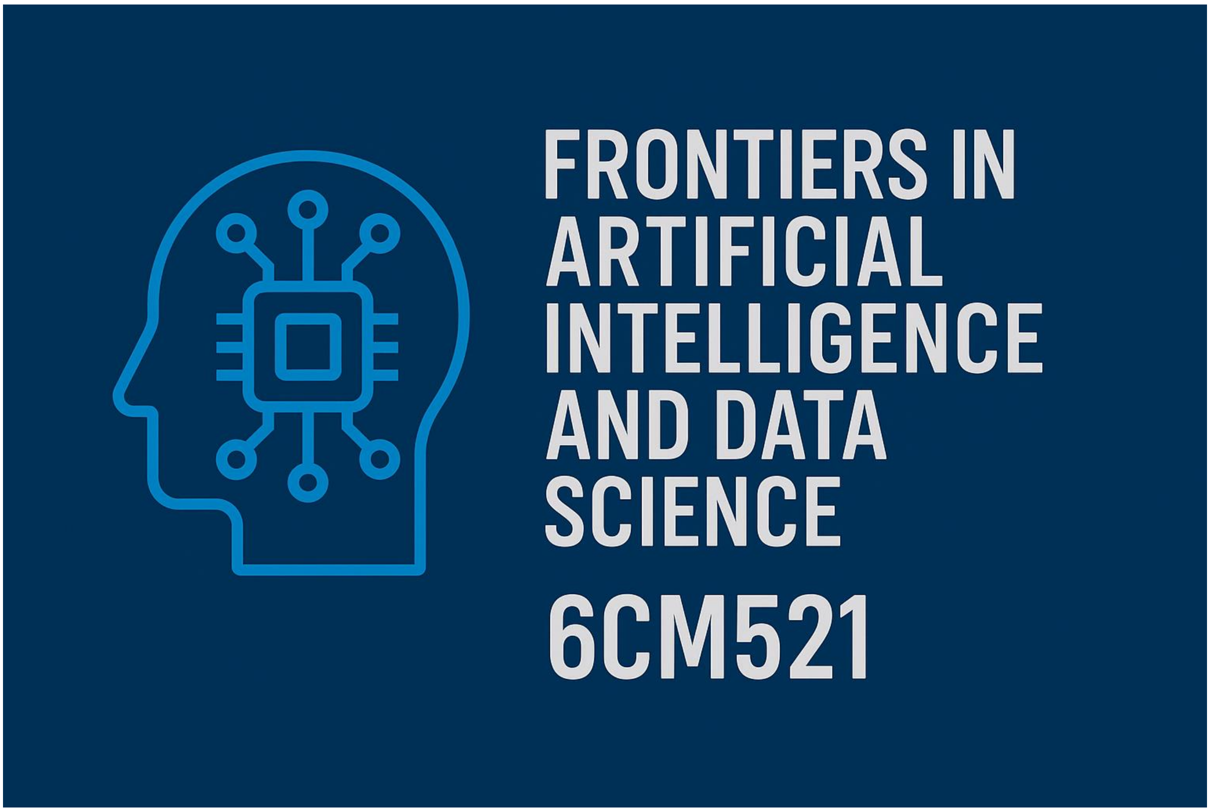
Frontiers in Artificial Intelligence and Data Science

6CM521



Assessment Brief

Associate Professor Harry Yu

Module Leader

- Harry Yu
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- Friday (10:00 – 11:00)

Key dates and details

Assessment Detail	Assessment Information
Assessment Mode:	Individual report/presentation and Artefact
Assessment Weighting:	100%
Word count/Length:	1000 words +/-5%.
Learning Outcomes:	1 and 2
Submission Method:	Blackboard Assignment and Panopto
Submission Date:	12:00, 28th Nov 2025 (Part A) and 1st May (Part B) 2026
Provisional Feedback Release Date:	**16:00, 12th Dec 2025 and 21st May 2026**

Description of the assessment

Integrated AI Capstone Project (Two-Part Individual Assessment)

Overall Aim:
Enable each student to develop a specialised AI component—spanning vision, audio, language, or learning algorithm—that will later integrate into a powerful, multi-modal AI agent with image processing & generation, object recognition, sound recognition & synthesis, intelligent communication, and continual learning.

Weighting:

- Part A (Preparation & Design): 30% of final grade
- Part B (Implementation, Integration & Evaluation): 70% of final grade

Relationship to Programme Assessment Strategy

This module is designed for all programmes that related to Computing subjects and details are defined to suit the purpose.

Attributes and Skills

	Skills	Links to useful resources
☒	Critical thinking	Develop@Derby Skill Set
☒	Communication	Develop@Derby Skill Set
☒	Collaboration	Develop@Derby Skill Set
☒	Creative problem solving	Develop@Derby Skill Set
☒	Self-direction & planning	Develop@Derby Skill Set
☒	Numeracy, statistics & financial literacy	Develop@Derby Skill Set
☒	Digital	Develop@Derby Skill Set
☒	Resilience	Develop@Derby Skill Set
☒	Adaptability	Develop@Derby Skill Set
☒	Leadership & future thinking	Develop@Derby Skill Set

Assessment Content

Part A: Preparation & Design

Objective:
Individually select a specific AI component within one of four domains, identify relevant datasets and technologies, and present a detailed design in a seminar session.

Deadline: Week 10

A. Topic Options (choose 1 topic and datasets are provided):

1. **Machine Vision for Object Recognition in Airport Security**

Scenario

Develop an object **recognition** ML or AI algorithm to identify prohibited/illegal items in airport security check context

Baggage screening (e.g., X-ray or RGB images of bags)

Focus strictly on **algorithmic object recognition** (detection/classification/segmentation as appropriate). **No** hardware, robotics control, or physical deployment is required.

Objectives

Your algorithm should:

- **Detect and classify** target item categories relevant to airport security (e.g., knives, guns, explosive components, scissors, batteries where restricted, liquids above allowance, etc.—choose a clear label set aligned with your dataset).
- Handle **occlusion, clutter, overlapping items, and class imbalance** common in security imagery.
- Demonstrate **robustness** via sensible data splits and at least one generalisation technique (e.g., augmentation, domain adaptation, or cross-dataset testing if feasible).

Constraints & Scope

- **Recognition only**: no multi-object tracking or robot control.
- You may use any modern CV approach (e.g., CNN/Transformer detectors, foundation models with fine-tuning, zero-shot with promptable vision models), but justify your choice.
- Clearly document your **label taxonomy**, **positive/negative classes**, and **handling of ambiguous items**.

2. **Large Language Model Base Dialogue System**

The task of this topic is to develop a **Large Language Model (LLM)-based dialogue system** designed for a customer support AI application. The goal of the project is to create a system capable of understanding customer queries, generating appropriate responses, and maintaining a coherent conversation flow.

For this assessment, you are **only required to focus on the algorithmic and system design aspects** of the **dialogue system**. You do **not** need to address deployment, integration with external databases, or hardware considerations.

Your work should demonstrate how your system can:

- Interpret and process customer queries using LLM-based techniques.
- Generate contextually appropriate, accurate, and clear responses.
- Manage conversational context to support ongoing dialogue.

B. Deliverables:

1. **Topic Selection & Justification** (10%)
    - One-page description of chosen AI component, relevance, and potential applications.
2. **Datasets & Technology Overview** (20%)
    - Critical analysis of the applied datasets (if you use any) in terms of suitability and limitations. You should use the datasets provided below (not prefer to use other datasets but if you like to use other datasets, then you need to discuss with your module leader beforehand to get an agreement. Otherwise, your work will not be marked as non-submission).

Machine Vision for Object Recognition for airport security (use only one of them)

1. **GDXray+ – domingo mery**
2. **OPIXray-author/OPIXray**

Large Language Model based Dialogue System

1. https://github.com/budzianowski/multiwoz (GitHub)
2. bitext/Bitext-customer-support-llm-chatbot-training-dataset · Datasets at Hugging Face

    - Review **three** key technologies or algorithms (2022–2025), summarizing strengths and limitations.
    - Provide a preliminary selection of development tools and platforms (e.g., TensorFlow, PyTorch, OpenCV).
    - **10-minute individual explanation** about above two points in 5 presentation slides.

C. Assessment Criteria (see marking rubric for detail):

- **Relevance & Scope:** Clear, focused component choice with compelling justification.
- **Dataset analysis:** Appropriateness and accessibility of datasets; understanding of data characteristics.
- **Technology Insight:** Depth of algorithmic/technical review and rationale for tool selection.

Part B: Implementation, Integration & Evaluation

Objective:
Summarising an investigation of the current cutting-edge AI framework to develop your selected AI module, integrate it into a master AI agent and a final demo.

The Master module will be provided by module leader

Deadline: Week 21

A. Deliverables:

1. **Prototype Implementation** (50% of Part B)
   - Working code for your AI component, with clear instructions for setup and execution.
   - Baseline performance metrics on test data.
2. **Capability of Integration** to real world system (10% of Part B)
   - Provide integration scripts or API wrappers, and a system overview document.
3. **Final Demo & Report** (10% of Part B)
   - **Demo (Max 10 min)**: Live demonstration showcasing

B. Assessment Criteria (see marking rubric for detail):

- **Functionality & Performance:** Easch module and the integrated system perform reliably with documented metrics.
- **Integration Robustness:** providing Seamless data flow, clear APIs, and error handling between modules.
- **Communication & Reflection:** Quality of demo presentation to show the understanding of the work and depth of analysis in the final report.

Assessment Rubric

Part A

Band	A1. Topic Selection & Justification (10%)	A2. Technology Review (10%)	A3. Technique & Platforms (10%)
90-100	Exceptionally clear, original and tightly scoped problem; compelling rationale linked to real use-case and LO's; anticipates risks/ethics and evaluation strategy.	Incisive, critical synthesis of ≥3 state-of-the-art methods (2022-2025) with evidence-based comparison; identifies gaps and motivates a preferred approach; exemplary referencing.	Toolchain is optimal and justified (perf., licensing, hardware, MLOps); shows feasibility (env/specs) and alternatives; integration path articulated.
80-89	Excellent focus and relevance; strong justification with evidence; clear success criteria.	Thorough, critical comparison of ≥3 methods; pros/cons, contexts, and metrics are well argued; very good sources.	Very strong, well-argued algorithm or model choices with constraints considered; clear setup details.
70-79	Clear and relevant scope; convincing justification; good link to outcomes.	Solid comparison of ≥3 methods with strengths/limits; minor gaps in depth or evidence.	Appropriate algorithm or model choices with sensible justification; notes on environment and data flow.
60-69	Good topic fit and rationale; some attention to risks/limits.	Competent review of 3 methods; mostly descriptive with some critique.	Reasonable tools; justification adequate but surface-level.
50-59	Satisfactory topic choice; rationale present but thin; aims somewhat generic.	Basic summary of 3 recent methods; limited critical insight or outdated sources.	algorithm or model choices listed with minimal justification/feasibility detail.
40-49	Marginal fit; weak justification; aims unclear/over-broad.	Superficial or <3 suitable methods; weak linkage to task; sparse/poor sources.	Algorithm or model choices with inappropriate or poorly matched; missing environment details.
30-39	Unclear/unsuitable topic; little justification; mismatched to brief.	Minimal, inaccurate, or largely off-topic review; little/no criticality.	Algorithm or model choices choices unjustified and impractical.
0-29	No credible topic or rationale.	No meaningful review.	No coherent tool plan.

Part B

Band	B1. Functionality & Performance (Prototype 50%)	B2. Integration Robustness (10%)	B3. Communication & Reflection (Demo + Report 10%)
90-100	Near state-of-the-art results on appropriate metrics (e.g., mAP/F1/WER/BLEU/latency), reproducible build (clear README, env file, seeds), strong testing (unit/integration), efficient use of compute; insightful error/ablation analysis.	Seamless pipeline integration with clean APIs, container/CI ready; excellent data flow, graceful error handling, config & logging.	Exceptional 8–10 min demo; compelling narrative with live metrics; report is publication-quality: rigorous method, limitations, ethics/safety, risks, and future work; immaculate referencing.
80-89	Excellent accuracy/robustness; reproducible; good test coverage; sensible baselines; thoughtful analysis of failures.	Very strong integration possibility; robust interfaces and monitoring; few minor issues only.	Excellent demo and report; clear insights, strong visuals; limitations well discussed.
70-79	Strong performance versus baselines; reproducible with minor gaps; sensible tests; reasonable efficiency.	Solid integration design; clear APIs; minor rough edges; adequate error handling.	Very good demo and report; coherent analysis; limitations acknowledged.
60-69	Good performance; builds and runs with clear instructions; some tests; basic analysis of results.	Works within the master pipeline; some minor fragility; basic logging/handling.	Good demo; report covers approach and results; limited critical reflection.
50-59	Meets core functionality; baseline metrics reported; setup mostly clear; minimal tests; little analysis.	Integrates at a basic level; manual steps or brittle parts; limited handling of edge cases.	Satisfactory demo; report descriptive; little discussion of limits/ethics.
40-49	Marginal: unreliable results; missing/unclear metrics; reproducibility doubtful.	Partial/fragile integration; unclear interfaces; frequent failures.	Weak demo; report lacks structure and analysis.
30-39	Major defects; fails key requirements; no credible evaluation.	Integration largely non-functional.	Ineffective demo; report inadequate.
0-29	No working prototype.	No integration achieved.	No meaningful demo/report.

Assessment AI-assistance

In this assessment AI-assistance is permitted in the following ways:

Searching related learning resources to support the development of the solution

Review on useful technologies and their implementations.

In this assessment AI-assistance should not be used for:

Generating any contents including codes, presentations and documents

Generation of solution design and datasets

It is YOUR RESPONSIBILITY to check ALL information generated by generative AI tools. Any misuse of generative AI tools could be considered ethical academic misconduct (as per Academic Regulations, Academic Misconduct, Section J2). If in doubt, please consult your module leader.

Anonymous Marking

Submissions in Blackboard

You must submit your work using your **student number** to identify yourself, not your name. You must not use your name in the text of the work at any point. When you submit your work in Turnitin you must submit your student number within the assignment document and in the *Submission title* field in Turnitin. Guidance is available showing how to do this.

Assessment Regulations

The University's regulations, policies and procedures for students define the framework within which teaching and assessment are conducted. Please make sure you are familiar with these regulations, policies and procedures.